

JAGANAATH A

Generative AI & Machine Learning Engineer

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TECHNICAL SKILLS

AI / ML Frameworks : LLMs, RAG, Prompt Engineering, LangChain, LangGraph, AI Agents, MCP, Multi-Agent Systems, Supervised Learning, OCR, NLP, Vector Databases (Chroma), Embeddings, SLM, Computer Vision

Languages & Data : Python, Pandas, NumPy, SQL

Cloud Platforms : Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure

Databases : MongoDB, Vector Databases

Tools & DevOps : FastAPI, Docker, REST APIs, Streamlit, Chainlit, Git

EXPERIENCE

AI Engineer — Tata Consultancy Services (TCS)

November 2024 – Present

Project: TAX MODERNIZATION | Client: Walmart

- Implemented an **OCR pipeline** to extract structured text from 500K+ product images, improving catalog attribute coverage and reducing manual data entry overhead.
- Orchestrated end-to-end ML pipelines on GCP and integrated an **LLM-as-a-judge** validation framework, reducing operational costs by 50%.
- Trained supervised ML models for large-scale product classification, cutting categorization time from 10 days to 1 day.

Project: OPS BUDDY | Client: Walmart

- Built an **NLP-driven Incident Chatbot** that automated 70% of internal employee interactions, reducing incident resolution time by half.
- Engineered a state-driven, **multi-agent LangGraph framework** with query classification and routing, cutting P95 ticket resolution time by 50% across 3 support categories.
- Implemented a **RAG pipeline with a vector database**, achieving 95% response accuracy for knowledge-based inquiries.
- Eliminated 40% of Tier-1 support overhead by building a RAG knowledge base over 1,000+ technical documents, enabling immediate and accurate employee self-service resolution.

Project: RIO – Retail Inventory Ordering | GenAI Agent-Based Automation Platform | Client: Starbucks

- Built AI agents to migrate **Microsoft Power Apps** to **React** by analyzing Power Apps source code and generating equivalent React layouts.
- Automated unit test generation for migrated React code, reducing **UI development effort** by 80%.
- Reduced the overall **Software Development Life Cycle (SDLC)** effort by 50% through end-to-end AI-driven automation of migration, code generation, and testing, enabling faster project delivery.

Project: PLATFORM INSIGHTS BOT — GenAI-Powered NL2SQL Assistant | Client: Walmart

- Developed an **end-to-end Text-to-SQL** engine achieving 80%+ accuracy across 10+ complex schemas through schema-aware generation and rule-based validation, reducing analysis time by 70%.
- Engineered an automated code-generation layer for **real-time KPI computation** across large-scale datasets, reducing ad-hoc query turnaround from hours to under 2 minutes.
- Integrated **NLG-driven summaries** and interactive visualizations into a single reporting dashboard, reducing time-to-insight for business teams.

EDUCATION

B.E. in Computer Science and Engineering

November 2020 – April 2024

Sri Sai Ram Institute of Technology | CGPA: 8.4

CERTIFICATIONS

- AWS Certified AI Practitioner
- AWS Certified Cloud Practitioner
- AWS Certified Machine Learning Engineer – Associate
- Microsoft Azure AI Fundamentals